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(54) **METHOD FOR SMOKING CESSATION**

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See application file for complete search history.

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(57)

ABSTRACT

A method of treating a subject for smoking cessation including periodically administering to the subject a pharmaceutically effective amount of cefdinir to treat the subject. Cefdinir for use in treating a subject for smoking cessation. A pharmaceutical composition including a pharmaceutically effective amount of cefdinir for use in treating a subject for smoking cessation.

3 Claims, No Drawings

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METHOD FOR SMOKING CESSATION**CROSS-REFERENCE TO RELATED APPLICATIONS**

The present application is a continuation-in-part of U.S. application Ser. No. 16/486,762, filed Aug. 16, 2019, which is a U.S. national phase application filed under 35 U.S.C. § 371 of International Application No. PCT/US2018/050404, filed Sep. 11, 2018, which claims priority to U.S. provisional application Ser. No. 62/560,959, filed Sep. 20, 2017, the disclosures of each of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

The present invention relates to methods for smoking cessation by administration of a pharmaceutically effective amount of cefdinir.

BACKGROUND OF THE INVENTION

Tobacco use causes nearly 6 million deaths per year globally, with current trends showing that tobacco use will cause more than 8 million deaths annually by 2030. In the United States alone, cigarette smoking is responsible for more than 480,000 deaths per year, including more than 41,000 deaths resulting from secondhand smoke exposure. The economic cost of smoking for the United States is more than \$300 billion each year, including nearly \$170 billion in direct medical care for adults and more than \$156 billion in lost productivity due to premature death and exposure to secondhand smoke.

At the present time, existing products for assisting in smoking cessation include nicotine-replacement products, which may take the form of a patch, gum, nasal spray, inhaler, or lozenge. Success rates relating to these products have been found to be as low as 15% after one year. Moreover, there are non-nicotine-based prescription medications on the market, such as varenicline (under the trade name of CHANTIX) and bupropion (under the trade name of ZYBAN). These medications have similarly low success rates, with CHANTIX being found in clinical trials as having only a 14% success rate after six months, while simultaneously resulting in tens of thousands of patient adverse reactions being reported to the Food and Drug Administration (FDA). Therefore, there is a need for effective and safe means for smoking cessation.

SUMMARY OF THE INVENTION

This invention provides a method of treating a subject for smoking cessation including periodically administering to the subject a pharmaceutically effective amount of cefdinir to treat the subject.

Implementations of the invention may include one or more of the following features. The pharmaceutically effective amount of cefdinir may be effective to reduce a clinical sign or symptom associated with tobacco addiction or smoking addiction in the subject, and/or a tobacco-induced health issue or a smoking-induced health issue in the subject. The periodic administration may be effected orally, and cefdinir may be administered in capsule form or oral suspension form. An amount of cefdinir administered may be in a range of 125-600 mg per dose, including 125 mg per dose, 250 mg per dose, 300 mg per dose, or 600 mg per dose. The periodic administration may be a once-daily administration or a

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twice-daily administration. The periodic administration may continue for at least 5 days or for at least 10 days. The subject may be human.

This invention also provides cefdinir for use in treating a subject for smoking cessation.

This invention further provides a pharmaceutical composition including a pharmaceutically effective amount of cefdinir for use in treating a subject for smoking cessation.

According to one approach, a pharmaceutical composition has a pharmaceutically effective amount of cefdinir for use in treating a subject for smoking cessation, and a fluid formulation. The composition may be configured to be aerosolized.

According to another approach, the composition may be configured to be vaporized and heat stabilized in the environment of vaporizer heating assemblies for electronic cigarettes to maintain pharmaceutically effectiveness. A ground Cefdinir powder may be suspended in a 0.001 g per 1 mL ratio (7.5%)-2.0 g per 1 mL ratio with the fluid formulation; and the fluid formulation may be a vaporizing base comprising at least one of glycerin and propylene glycol. The fluid formulation may be in a ratio of roughly 92% of volume in relation to the cefdinir and any other components. Other components include a fluid flavoring of the composition at 0.5% of the total composition volume.

According to another approach, a pharmaceutical composition may have a pharmaceutically effective amount of cefdinir for use in treating a subject for smoking cessation, and a vehicle to deliver the Cefdinir to a patient selected from the list of a transdermal patch, liquid drops, oral pump spray, melting paper, chewable tablets, lozenges, gum, and the like, and combinations thereof.

DETAILED DESCRIPTION OF THE INVENTION

This invention provides a method of treating a subject for smoking cessation including periodically administering to the subject a pharmaceutically effective amount of cefdinir to treat the subject.

In one embodiment, the pharmaceutically effective amount of cefdinir is effective to reduce a clinical sign or symptom associated with tobacco or smoking addiction and/or tobacco- or smoking-induced health issues in the subject.

In one embodiment, the periodic administration of cefdinir is effected orally. In one embodiment, cefdinir is administered in capsule form. In another embodiment, cefdinir is administered in oral suspension form.

In one embodiment, the amount of cefdinir administered is 125-600 mg per dose. In another embodiment, the amount of cefdinir administered is 125 mg per dose. In another embodiment, the amount of cefdinir administered is 250 mg per dose. In another embodiment, the amount of cefdinir administered is 300 mg per dose. In another embodiment, the amount of cefdinir administered is 600 mg per dose.

In one embodiment, a dose or dosage of cefdinir is administered to the subject once per day. In another embodiment, a dose or dosage of cefdinir is administered to the subject twice per day.

In one embodiment, the periodic administration continues for at least 5 days. In one embodiment, the periodic administration continues for 5 days. In another embodiment, the periodic administration continues for 10 days.

In one embodiment, the subject is human.

This invention also provides cefdinir for use in treating a subject for smoking cessation.

This invention further provides a pharmaceutical composition including a pharmaceutically effective amount of cefdinir for use in treating a subject for smoking cessation.

For the foregoing embodiments, each embodiment disclosed herein is contemplated as being applicable to each of the other disclosed embodiments.

Disclosed is a method of treating a subject for smoking cessation using cefdinir. Cefdinir is a third-generation oral cephalosporin antibiotic commonly sold under the trade names of CEFZON or OMNICEF. It has been previously understood that cefdinir may be used to treat various types of bacterial infections, such as bronchitis (infection of the airway tubes leading to the lungs), pneumonia, and infections of the skin, ears, sinuses, throat, and tonsils. The effects of cefdinir on smoking cessation have not been reported. As described herein, administration of cefdinir is effective to treat a subject for smoking cessation.

Administration of cefdinir is advantageous over existing treatments for smoking cessation because cefdinir can be administered orally and has well-documented success in terms of safe administration with minor side effects.

As used herein, and unless stated otherwise, each of the following terms shall have the definition set forth below.

As used herein, an "amount," "dose," or "dosage" of cefdinir as measured in milligrams refers to the milligrams of cefdinir present in a preparation, regardless of the form of the preparation. Therefore, a "dose of 125 mg cefdinir" means the amount of cefdinir in a preparation is 125 mg, regardless of the form of the preparation.

As used herein, "a subject" generally means a subject who is attempting to quit smoking or using tobacco-based products, such as cigarettes.

As used herein, "effective" or "pharmaceutically effective" when referring to an amount of cefdinir refers to the quantity of cefdinir that is sufficient to yield a desired therapeutic response without undue adverse side effects (such as toxicity, irritation, or allergic response) commensurate with a reasonable benefit/risk ratio when used in the manner of this invention.

As used herein, "treating" encompasses, e.g., inducing inhibition, regression, or stasis of a disorder, or lessening, suppressing, inhibiting, reducing the severity of, eliminating, or ameliorating a symptom of the disorder, where the disorder includes tobacco or smoking addiction, tobacco- or smoking-induced health issues, and the like.

In an embodiment of the present invention, a subject is administered a pharmaceutically effective amount of cefdinir at several times during a treatment regimen. At the conclusion of this treatment regimen, the subject feels no desire or, at a minimum, a decreased desire to smoke or otherwise use tobacco-based products, such as cigarettes.

Table 1 illustrates cefdinir plasma concentrations and pharmacokinetic parameter values following administration of single 300 and 600 mg oral doses of cefdinir to adult subjects.

TABLE 1

Mean ($\pm$ SD) Plasma Cefdinir Pharmacokinetic Parameter Values Following Administration of Capsules to Adult Subjects			
Dose	C _{max} (μ g/mL)	t _{max} (hr)	AUC (μ g $\cdot$ hr/mL)
300 mg	1.60 (0.55)	2.9 (0.89)	7.05 (2.17)
600 mg	2.87 (1.01)	3.0 (0.66)	11.1 (3.87)

As indicated above, according to one approach the Cefdinir may be administered in a suspended form in, for example, an aqueous medium. For exemplary purposes, such formulations may include a variety of components safe for human use including but not limited to flavorings (e.g., mint, eucalyptus, popular vaping flavorings, etc.), nutrients, caffeine, and the like and combinations thereof. The formulations may be configured for delivery to the human through oral suspensions; vaporized ('vape') form (e-cigarette), oral inhaler, and nasal inhaler. A vehicle to deliver the Cefdinir to a patient may be selected from the list of a transdermal patch, liquid drops, oral pump spray, melting paper (such as made from cellulose gum or like those sold under the tradename LISTERINE TABS), chewable tablets, lozenges, gum, and the like, and combinations thereof.

For aerosolized versions for pulmonary delivery of the human, the dose amounts of Cefdinir may be similar to tablet or capsule form described above or configured to be a pharmaceutically active dose via the mode of transmission or delivery into the body. Similarly, for transdermal patches the dose would be calibrated to rate and amount of absorption.

In each instance, the formulations would need to be adjusted to meet the needs for the mode of delivery to maintain effectiveness. For example, for vaping or pulmonary delivery, the compositions would be sterile and balanced to the body pH at the point of delivery.

In one embodiment for vaping, for example, a vaping juice ('e-liquid') containing Cefdinir may include water, flavoring, propylene glycol, vegetable glycerin, and the like and combinations thereof. The vaporizing base may be propylene glycol, vegetable glycerin, or mixture of the two.

In one embodiment of the present invention, ground Cefdinir powder is presented in a 0.001 g per 1 mL ratio (7.5%)-2.0 g per 1 mL ratio with a vaporizing base. In one embodiment the vegetable glycerin and/or propylene glycol mixture is in a ratio of roughly 92% in relation to other components. In one embodiment of the present invention a fluid flavoring of the e-liquid mixture may be added at 0.5% of the total e-liquid mixture.

The embodiments and examples above are illustrative, and many variations can be introduced to them without departing from the spirit of the disclosure or from the scope of the appended claims. For example, elements and/or features of different illustrative and exemplary embodiments herein may be combined with each other and/or substituted with each other within the scope of this disclosure. The objects of the invention, along with various features of novelty which characterize the invention, are pointed out with particularity in the claims annexed hereto and forming a part of this disclosure.

What is claimed is:

1. A pharmaceutical composition comprising:

a pharmaceutically effective amount of ground cefdinir powder for use in treating a subject for smoking cessation,

a fluid formulation; and

a vehicle to deliver the pharmaceutical composition,

wherein ground cefdinir powder is suspended in the fluid formulation in a ratio of a 0.001-2.0 g cefdinir per 1 mL of the fluid formulation;

wherein the fluid formulation is a vaporizing base comprising at least one of glycerin and propylene glycol;

wherein the effective amount of cefdinir is 300-600 mg per dose and resulting in a subject plasma cefdinir pharmacokinetic value between 1.60 μ g/mL and 2.87 μ g/mL; and

wherein the composition is sterile and has a pH of a body pH of the subject at a point of delivery.

2. A pharmaceutical composition comprising:

a pharmaceutically effective amount of ground cefdinir powder for use in treating a subject for smoking cessation,

a fluid formulation;

wherein ground cefdinir powder is suspended in the fluid formulation configured for oral once-daily administration for at least 5 days in a ratio of a 0.001-2.0 g cefdinir per 1 mL of the fluid formulation,

wherein the composition is configured to be vaporized and heat stabilized in a vaporizer heating assembly of an electronic cigarette in a predetermined amount for the cefdinir to achieve pharmaceutical effectiveness,

wherein the effective amount of cefdinir is 300-600 mg per dose and resulting in a subject plasma cefdinir pharmacokinetic value between 1.60 µg/mL and 2.87 µg/mL; and

wherein the composition is sterile and has a pH of a body pH of the subject at a point of delivery.

3. A method of treating a subject for smoking cessation comprising periodically administering to the subject a pharmaceutically effective amount of the pharmaceutical composition of claim 1 to treat the subject.

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